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BIOPHYSICS

Tubulin polymerization dynamics are influenced by magnetic isotope effects consistent with the radical pair mechanism

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Weak magnetic fields have been shown to influence biological processes; however, the underlying mechanisms remain unknown as the energies involved are far below thermal energies challenging classical explanations. Microtubule cytoskeletal fibers offer an ideal system to test weak magnetic field effects due to their self-assembling capabilities, sensitivity to magnetic fields, and their central role in cellular processes. In this study, we use a combination of experiments and simulations to explore how nuclear spin dynamics affect microtubule polymerization by examining interactions between magnesium isotope substitution and weak magnetic fields. Our experiments reveal an isotope-dependent effect explicitly arising from nuclear spin properties. This nuclear spin-driven effect is enhanced under an applied weak 3-millitesla magnetic field. Our theoretical radical pair model achieves quantitative agreement with our experimental observations. These results support a connection between quantum spin dynamics and microtubule assembly, providing further insights into how weak magnetic fields may influence biomolecular functions.

INTRODUCTION

Weak magnetic fields and isotope effects have been observed to influence biological systems, yet the underlying mechanisms remain debated. Biological processes are generally understood through biochemical and thermodynamic principles; however, some reactions demonstrate sensitivities that classical models cannot easily explain—especially given that the corresponding energies from such weak magnetic fields are far below thermal energies at physiological temperatures. Notably, these effects have been reported across diverse systems, including electron transfer in cryptochrome (*1*, *2*), stem cells (*3*), neurogenesis (*4*), axon growth (*5*), the circadian clock (*6*), and cellular autofluorescence (*7*). These observations suggest that certain biochemical processes may depend on quantum properties such as electron and nuclear spin dynamics (*8*). At the same time, contradictory findings have been reported [e.g., no magnetic field effects on *Drosophila* behavior (*9*) and failure to reproduce magnetic sensitivity of cellular autofluorescence (*10*)], underscoring the need for careful controls and standardized methodologies. Consequently, rigorous experimental validation of quantum mechanical mechanisms in biologically relevant contexts remains essential.

A leading explanation for these effects is the radical pair mechanism (RPM) (*8*, *11*, *12*). The RPM involves transient pairs of atoms or molecules each with an unpaired electron (i.e., radicals) that are formed during chemical reactions (Fig. 1A). These radical pairs can exist in two distinct quantum states: a singlet state, where the electron spins are entangled and their combined spin is zero, and a triplet state, where the spins are aligned in a way that gives a total spin of one. Because of the interaction of the electron spins with the environment, external magnetic fields, and nuclear spins, these radical pairs can oscillate between singlet and triplet spin states (*11*, *13*). The rate at which a radical pair recombines to complete a chemical reaction depends on its spin state. This spin-dependent recombination underlies magnetically sensitive spin chemistry, explaining how weak magnetic fields influence biochemical processes (*14*, *15*). Isotopes of elements such as lithium, magnesium, calcium, or zinc can influence this process through hyperfine coupling, which involves interactions between their nuclear spins and electron spins. This coupling establishes a clear connection between spin-dependent isotope effects and the sensitivity of biological systems to weak magnetic fields (*8*).

In a biological context, flavin-tryptophan radical pairs in cryptochrome have been shown to generate magnetic field effects (*1*). Beyond this specific case, the RPM can provide a general framework for explaining weak magnetic field effects across diverse biological systems. However, many of these systems lack cryptochrome or even flavin, and some also exhibit isotope effects that remain poorly understood (*8*). Testing whether RPM-based mechanisms are at play in these contexts requires carefully designed experiments in other established molecular systems. Observation of combined weak field and isotope effects consistent with theoretical predictions would be a strong indication of radical pair involvement in biology. To our knowledge, no experiment has been able to conclusively show both weak magnetic field and isotope effects in a biologically relevant chemical system that is consistent with a theoretical description of radical pair dynamics. That is the objective of this investigation.

Microtubules offer an ideal biochemical system for assessing the broader applicability of the RPM beyond well-studied systems like bird navigation and cryptochrome. Microtubules are dynamic protein polymers essential for cell structure, intracellular transport, and cellular organization (*16*–*18*) in all eukaryotic systems. Their rapid polymerization and depolymerization enable cytoskeletal reorganization during processes like cell division and neuronal growth (*19*–*21*), making them targets for treating cancer and neurodegenerative

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diseases (*19*). Known factors that regulate microtubule polymerization dynamics include magnesium (Mg), pH, and temperature, as well as removal of the weak geomagnetic field (GMF) (25 to 65 μT) (*22*). How small changes in magnetic fields influence microtubule polymerization dynamics remains unknown. However, a recent theoretical investigation has proposed that spin dynamics can directly influence microtubule density through effects on polymerization rates (*23*). This RPM-based model has theoretically reproduced the experimentally observed hypomagnetic field (HMF) (<5 μT) effects on microtubule density (*22*) and further predicts isotope-dependent and applied weak magnetic field effects (*23*).

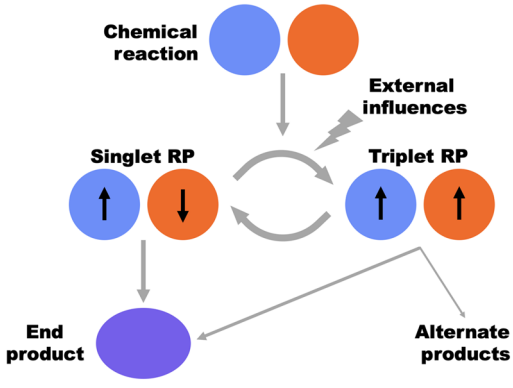


Fig. 1. Radical pair mechanism. A transient radical pair (RP) can be formed during a chemical reaction, which may interconvert between singlet and triplet states (arrows indicate the electron spin) depending on external influences such as magnetic fields or nearby nuclear spins. End products may be formed from the singlet or triplet state at different rates, whereas the triplet state may also lead to alternate chemical products.

In this study, we investigate the influence of weak magnetic fields and different Mg isotopes on microtubule polymerization dynamics. Mg, essential for microtubule dynamics, has been reported to play a role in spin-selective phosphorylation sensitive to weak fields and isotope substitution (*24*–*29*). The difficulty in reproducing magnetic field and isotope effects with Mg requires careful, controlled testing (*24*, *30*, *31*). As such, we measure changes in optical density (OD) to assess microtubule polymerization in the presence of Mg isotopes with and without nuclear spins in the absence or presence of a weak magnetic field while accounting for confounding factors such as pH, temperature, and differences in isotope weight. We report experimental evidence demonstrating substantial changes in microtubule polymerization dynamics induced by ^{25}Mg in the presence of a weak external magnetic field. We further show that a general RPM model is capable of reproducing experimentally observed Mg isotope and weak magnetic field effects on microtubule polymerization.

RESULTS

Weak magnetic field enhances tubulin polymerization in the presence of the ^{25}Mg isotope

To investigate the role of nuclear spin in microtubule polymerization, we measured the effect of Mg isotopes on tubulin assembly dynamics, both in the presence and absence of a uniform magnetic field of 2.99 ± 0.02 mT applied using a Helmholtz coil (Fig. 2A). In-house made polymerization buffers were all made with identical pH and shown to reproduce standard manufacturer polymerization curves (fig. S1). The high enrichment values of the isotopes ensured the minimal presence of paramagnetic ions (cobalt, copper, iron, manganese, molybdenum, vanadium, etc.) at concentrations less than 0.6 μM (figs. S2 and S3). Analysis of tubulin polymerization as measured by OD at 355 nm indicated that there was a significant enhancement in

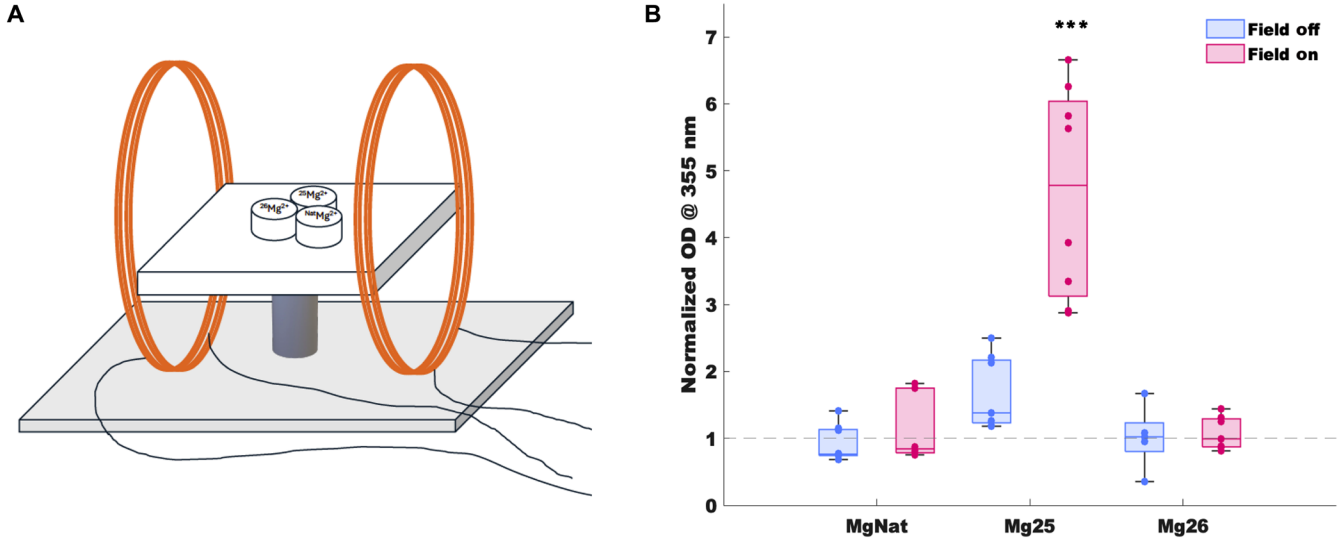


Fig. 2. Isotope effect on tubulin polymerization in the absence and presence of a 3-mT magnetic field. (**A**) Schematic representation of the magnetic field setup using Helmholtz coils. Tubulin samples containing ^{Nat}Mg(79% ^{24}Mg, 10% ^{25}Mg, 11% ^{26}Mg), ^{25}Mg, and ^{26}Mg were placed at the center of the Helmholtz coils, which generated a uniform magnetic field of 2.99 ± 0.02 mT (field on). (**B**) Box plots showing the median line, upper and lower quartile box, maximum and minimum whiskers, and scatter of individual replicates for the final OD measure of tubulin polymerization for ^{Nat}Mg, ^{25}Mg, and ^{26}Mg in the absence (blue) and presence (magenta) of the applied magnetic field of 2.99 ± 0.02 mT. OD values are in arbitrary units relative to the final values of tubulin polymerization with ^{26}Mg in the absence of an applied magnetic field. ***$P < 10^{-7}$, difference from ^{25}Mg in the absence of an applied field as determined by a two-way ANOVA with a Tukey-Kramer post hoc test. Notably, neither ^{Nat}Mg nor ^{26}Mg show a significant difference between field-on and field-off conditions.

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total tubulin polymerized ($P < 10^{-7}$) in the presence of ^{25}Mg compared to ^{Nat}Mg and ^{26}Mg with the application of the magnetic field (Fig. 2B).

To directly test alternative hypotheses that could explain the effects observed in Fig. 2, we specifically examined whether differences in isotope mass, well-to-well temperature differences, or well-to-well magnetic field differences could account for the observed results (fig. S5, B and C). No differences were observed to be due to increasing isotope mass (i.e., kinetic isotope effects) as ^{25}Mg is both lighter than ^{26}Mg and heavier than ^{Nat}Mg, which is primarily composed of ^{24}Mg (79%). Analysis of well-to-well temperature variations did not show a significant correlation between the OD and temperature for any condition ($P > 0.13$) and did not show a significant contribution to the observed effect ($P > 0.26$) (fig. S4A). Analysis of well-to-well magnetic field variations also did not show a significant correlation between the OD and magnetic field for either the field-off ($P > 0.11$) (fig. S4B) and field-on ($P > 0.19$) conditions (fig. S4C). All data, including raw and normalized polymerization data, temperature, and magnetic field measures, and statistical analyses, are provided in the Supplementary Materials.

A radical pair model of magnetic field and isotope effects on microtubule dynamics

To explain our experimental observations, we used a simple ordinary differential equation model of microtubule dynamics (*32*) describing the dynamic exchange between free tubulin molecules (*Tu*) and tubulin polymerized into microtubules (*MT*) such that the time (t)–dependent concentration of *MT* is defined as

$$[MT(t)] = \frac{k_p[P]}{k_p + k_d}[1 - e^{-(k_p+k_d)t}] \quad (1)$$

where k_p is the rate of polymerization, k_d is the rate of depolymerization, and $[P]$ is the concentration of total tubulin protein {i.e., $[P] = [MT] + [Tu]$}. As Mg is required for tubulin to hydrolyze guanosine triphosphate (GTP), and removal of the GTP cap increases microtubule depolymerization, we incorporate Mg-dependent magnetic field and isotope effects into the model by modulating k_d through changes in the radical pair triplet yield as follows

$$k'_d \propto \frac{\Phi'_T}{\Phi_T} k_d \quad (2)$$

where, k'_d, Φ_T, and Φ'_T represent the modified depolymerization rate constant, the triplet yield under the standard ambient magnetic field (i.e., GMF) with a zero spin Mg isotope, and the triplet yield arising from an applied magnetic field and hyperfine interactions due to a nonzero isotope nuclear spin, respectively. The triplet yield depends on the interaction terms, their spin relaxation rates, and the reaction rates for the radicals in the singlet and triplet states. Complete details concerning the model terms and parameters are in the Materials and Methods section.

To incorporate the magnetic field and Mg isotope effects, we use a generic RP model. In this model, each radical (A and B) is coupled to a spin-1/2 nucleus with hyperfine coupling constants (HFCCs) of $a_{A/B}$, by additionally including ^{25}Mg coupling to electron B with the HFCC of a_{25}. Electron A/B is subject to a spin relaxation rate $r_{A/B}$. The singlet reaction rate is k_S, and the triplet reaction rate is k_T.

Here, we focus on Zeeman and hyperfine interactions, which are not distance dependent. In our model, we consider the case that spin correlations arise from encounters between previously uncorrelated radicals A and B (i.e., an F-pair formation mechanism) (*33*). Using the differential evolution algorithm (*34*), the optimization of parameter values was carried out considering the following parameter bounds: r_A and $r_B \in [10^5, 10^7]$ 1/s, k_S and $k_T \in [10^4, 10^8]$ 1/s, a_A and $a_B \in [-5, 5]$ mT, and $a_{25} \in [-20, 20]$ mT.

Experimental results show that using ^{25}Mg (nuclear spin $I = 5/2$) with the applied 3-mT field significantly increases microtubule density compared to ^{25}Mg at the GMF and zero nuclear spin isotopes ($I = 0$ effectively for ^{Nat}Mg and ^{26}Mg) at either 3 mT or the GMF. In contrast, no significant change in microtubule density is observed between the zero spin isotopes for the 3-mT field and GMF conditions, indicating that the model must also account for these cases where the nuclear spin and magnetic field variations do not influence the outcome significantly. Furthermore, in the absence of a magnetic field (HMF ~ 0 mT), the microtubule density with ^{Nat}Mg is ~40% of that reported in the GMF (*22*). To model these behaviors, we compare two cases: Mg ions with nuclear spin $I = 5/2$ and those with $I = 0$. Using Eqs. 1 and 2 along with optimized parameters, the model successfully captures the field dependence of microtubule polymerization (Fig. 3A). The optimization process produced a varied array of parameter values that successfully reflect the observed behavior under different conditions. The recombination rates were optimized to span from $k_{rA} = 10^5$ to 4.1×10^5 s^{-1} and $k_{rB} = 1.32 \times 10^5$ to 2.6×10^6 s^{-1}. The reaction rates for singlet and triplet states ranged from $k_S = 6.1 \times 10^7$ to 9.4×10^7 s^{-1} and $k_T = 1.9 \times 10^5$ to 9.3×10^5 s^{-1}, respectively. The HFCCs were fine-tuned with a_A varying between 0 and 0.03 mT, a_{25} between 1.8 and 6.6 mT, and a_B between 0.9 and 4.2 mT (see Fig. 3A). The corresponding data file is available in the Supplementary Materials.

This model not only aligns with our experimental results but also offers predictive power for future studies. Specifically, the model predicts that microtubule polymerization will be further reduced at magnetic field strengths above 5 mT compared to the levels observed at 3 mT (see Fig. 3B). In addition, it indicates that polymerization significantly decreases under HMF conditions compared to GMF conditions (see Fig. 3C).

DISCUSSION

Our experiments reveal a Mg isotope–dependent magnetic field effect on microtubule polymerization. Having ruled out influences from pH, temperature variations, the kinetic isotope effect, well-to-well magnetic field variations, and paramagnetic impurities, we conclude that the observed effect arises from the applied weak magnetic field and the nonzero nuclear spin of ^{25}Mg (Fig. 2). These results align with an RPM description of isotope and weak magnetic field effects, strongly suggesting that an RPM modulates biochemical reaction rates during microtubule polymerization (*23*, *35*).

Although our experiments indicate a role for Mg in the observed results, the RPM model presented here is general in nature and does not explicitly identify the radicals involved. Rather, it incorporates isotope-specific effects by including two generic spin-1/2 nuclear spins, one associated with each radical, as well as an additional ^{25}Mg nuclear spin coupled to one of them. Here, ^{25}Mg is not considered one of the radical partners but considered to influence one of the nearby radicals via its nuclear spin. Parameters for this generic

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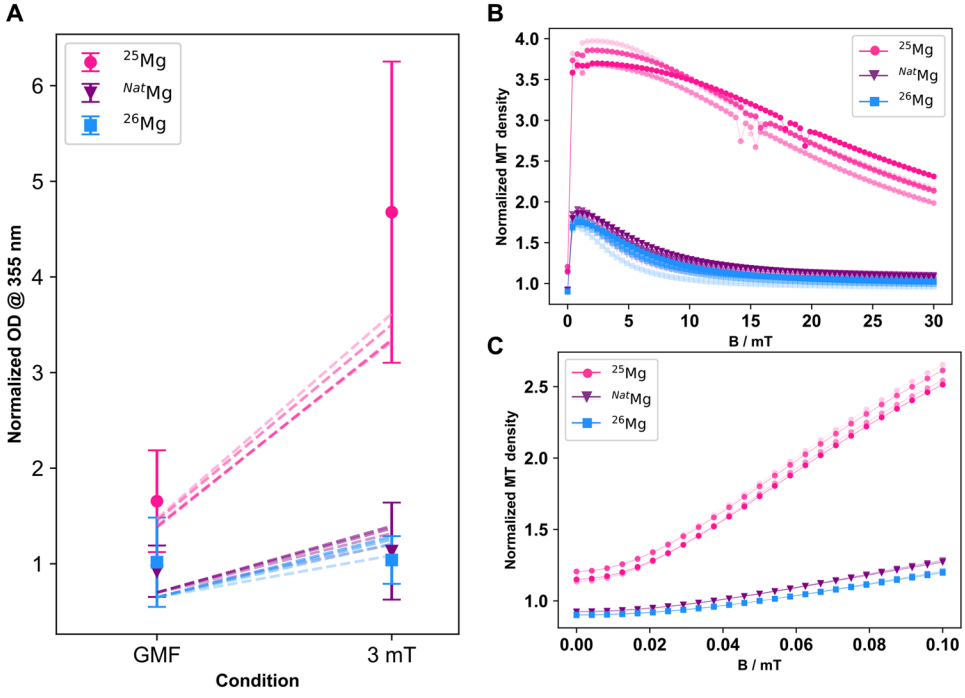


Fig. 3. Modeling magnetic field and isotope effects on microtubule dynamics. (**A**) Comparison between the experimental data and radical F-pair model for microtubule density using ^{25}Mg (magenta), ^{26}Mg (blue), and ^{26}Mg (purple) under two conditions with exposure of 3 mT and the GMF of 0.05 mT. Experimental data are presented as means ± SD. Dashed lines connect model-predicted points at 0 and 3 mT for visualization only and do not indicate intermediate model values. The optimization procedure yielded a diverse set of parameter values that effectively capture the observed behavior across multiple conditions as indicated by the various dashed plots. The optimized values for the recombination rates were found to range from $k_{rA} = 1.1 \times 10^5$ to 1.4×10^5 s^{-1} and $k_{rB} = 1.7 \times 10^5$ to 4.9×10^5 s^{-1}. The singlet and triplet reaction rates varied between $k_S = 8.3 \times 10^7$ and 9.2×10^7 s^{-1} and $k_T = 6.6 \times 10^5$ to 8×10^5 s^{-1}, respectively. The HFCCs were optimized as $a_A = 0.01$ mT, a_{25} from 5.3 to 6.3 mT, and a_B from 1.4 to 2.6 mT. This range of parameters are used in (B) and (C) as well. (**B**) Microtubule polymerization will be further reduced with ^{25}Mg at magnetic field strengths above 5 mT, compared to the levels observed at 3 mT. (**C**) Normalized microtubule density decreased under HMF conditions (<5 μT). In the classical microtubule model, the polymerization rate is $k_p = 5 \times 10^{-5}$ s^{-1}, the depolymerization rate is $k_d = 3 \times 10^{-3}$ s^{-1}, and the initial tubulin concentration is 3 g/liter.

model were optimized in an unbiased way using the differential evolution algorithm to best capture the observed behaviors (Fig. 3). Although general in nature, the parameters identified can provide insights on the potential radical pairs involved. First, the initial spin state of a radical pair can provide information on the radicals involved. As the radicals are not predefined in our model, we presented modeling results based on a fully mixed initial state. Both purely triplet and singlet initial states for the RPs in our analysis were also explored, with only triplet initial states being able to reproduce the experimental results (see the Supplementary Materials). Parameter optimization within our generic radical pair model further indicated that the unpaired electron on the radical opposite the Mg-bound species must be coupled to its associated nuclear spin with a very weak HFCC (Fig. 3).

All interactions involved in tubulin self-assembly are noncovalent molecular interactions except for the Mg-dependent biochemical hydrolysis of GTP. As such, the chemical intermediates of this process are the ideal candidates to be involved in an RPM. Although the exact chemical mechanism of GTP hydrolysis remains actively under investigation (*36*, *37*), it broadly involves Mg coordinating a water molecule to initiate a nucleophilic attack on the terminal γ-phosphate group of GTP. This results in cleavage of the phosphate bond and formation of guanosine diphosphate (GDP) and inorganic phosphate (P_i). During this hydrolytic process, the coordinated water molecule is cleaved into hydroxyl (OH) and proton (H^+) ions, which subsequently neutralize the free phosphate group, finalizing the biochemical reaction. The GTP hydrolysis mechanism then suggests four possible candidates for the RPM: GDP, P_i, OH, and Mg. Prior works have proposed a $Mg^{\bullet+}$-P_i radical pair (*24*–*29*), particularly in adenosine triphosphate (ATP)–associated P_i groups and protein kinase–mediated P_i transfer via an ion-radical mechanism. However, studies suggest that, although $Mg^{\bullet+}$ radicals are chemically feasible and observed in specific nonbiological contexts (*38*), their formation in standard biological systems, such as microtubule polymerization, have no experimental confirmation. In addition, at the interatomic distance between $Mg^{\bullet+}$ and P_i on GTP-tubulin, the strong exchange interaction dominates, leading to rapid recombination or suppression of magnetic field sensitivity. $Mg^{\bullet+}$ is also highly unstable in aqueous solution and seems unlikely to participate in biologically meaningful radical pairs (*30*). This is supported by our model. The $Mg^{\bullet+}$ radical is known to have a very large HFCC (~20 mT) (*39*). The search space parameter used for the Mg HFCC in our optimization covered −20 to 20 mT, accommodating the $Mg^{\bullet+}$ radical HFCC strength. However, the resulting optimal range of 1.8 to 6.6 mT is much less. This effective HFCC supports that the unpaired electron of the relevant nearby radical is delocalized and weakly coupled to the nuclear spin of ^{25}Mg. Thus, at first approach, $Mg^{\bullet+}$ and P_i on the GTP appear as unlikely radical candidates to explain the observed magnetic field effects.

Keeping this in mind, it must be noted that, in the present work, we focused on distance-independent Zeeman and hyperfine interactions,

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neglecting the distance-dependent exchange and dipolar interactions. Previous studies have suggested that dipolar and exchange interactions could neutralize each other at specific distances (*40*). More recently, a triad radical model has been proposed as an effective method to mitigate the challenges posed by dipolar interactions (*33*). In practice, radical pairs are likely dynamic entities, frequently moving closer and farther apart. Such movement allow for variance in dipolar interactions, which are averaged out over time, while maintaining sufficient distance most of the time to minimize exchange interactions. This dynamic nature ensures that radical pairs occasionally approach closely enough to permit chemical reactions while remaining sufficiently separated otherwise to preserve sensitivity to weak magnetic fields. Addressing this balance is a general challenge within the RPM, one that we acknowledge but do not explore in depth here. Following recent works on biradical flavin adenine dinucleotide (*41*–*46*), these dynamics can be further investigated in subsequent studies to provide information on potential radicals involved in the current microtubule system.

Beyond the GTP hydrolysis mechanism we need to consider other potential sources for potential radicals. The experimental biochemical assays contain the buffer components Pipes, EGTA, and glycerol. Pipes is an ethanesulfonic acid buffer used to maintain a stable pH, whereas EGTA is a calcium-selective chelating agent commonly used in biological assays to control or suppress calcium ion concentration due to calcium's known depolymerizing effect on microtubules. Glycerol, a sugar alcohol included to stabilize microtubules, mimics the intracellular environment and contributes to maintaining microtubule integrity. Regarding radical formation, none of these buffer components are conventionally recognized to directly participate in, or substantially facilitate, radical formation under typical experimental conditions.

To clarify the underlying chemistry in the RPM model, theoretical approaches could include quantum chemical modeling of radical formation pathways, particularly focusing on the stability and reactivity of Mg or OH radicals in the tubulin microenvironment. Experimentally, electron paramagnetic resonance spectroscopy could be used to detect and characterize short-lived radical intermediates during microtubule assembly, providing direct evidence of their existence and magnetic properties. Prior research has proposed a tryptophan-superoxide radical pair affecting microtubule polymerization (*23*) to explain HMF effects on microtubules (*22*), but the enzymatic GTPase process of tubulin, which involves the hydrolysis of GTP, is not known to directly produce superoxide, and superoxide is not specifically introduced into our biochemical assay. Photoinduced radical pairs of superoxide with tryptophan, tyrosine, or GTP products remain a possibility because our measurements were conducted under ambient light. However, organic radicals with the properties needed to show magnetic field effects usually have delocalized unpaired electrons and many hyperfine couplings, which suppress weak-field magnetic effects (*47*, *48*). Appreciable sensitivity occurs mainly when one partner has very few couplings, which is a rare biological circumstance (*47*, *49*). Candidates, such as superoxide or nitric oxide, relax too quickly to matter except under special conditions (*47*, *50*, *51*). Even so, the potential for indirect radical generation, particularly via interactions with dissolved oxygen or other reactive species, cannot be entirely excluded and merits further investigation to explain the observed effects. Further experiments under deoxygenated conditions or under dark conditions would help rule out radical pairs from autoxidation or photoinduction.

A broader objective is to develop a comprehensive model that fully explains the observed Mg isotope and magnetic field effects, including their implications for weak, low-frequency magnetic fields (*52*) in biological systems. Although our current framework captures key aspects of isotope-dependent radical pair dynamics in microtubules, further refinement is needed to integrate findings from in vivo experiments, where additional biochemical complexity and environmental factors may influence these effects. This is of key importance to the broader applicability of the observed results. For example, disruptions in microtubule dynamics are closely linked to neurodegenerative diseases like Alzheimer's disease (AD) (*53*). Notably, elevated brain Mg has been shown to exert synaptoprotective effects in AD models (*54*), whereas weak magnetic fields are shown to influence learning and memory (*4*). As our findings show a weak magnetic field influence on microtubule polymerization via a Mg-sensitive RPM, this suggests that external magnetic fields or isotopic variations may affect neuronal stability via a quantum-sensing mechanism. Investigating these quantum effects may open avenues for understanding AD progression and developing therapeutic strategies.

In conclusion, the results presented here show weak magnetic field and magnesium isotope effects on microtubule polymerization dynamics that are consistent with a theoretical description of radical pair dynamics. The general RPM presented herein accurately explains the isotope effects in microtubule assembly, despite the underlying chemistry remaining unclear. This demonstrates the broad potential of the RPM for understanding weak magnetic field effects and isotope effects in biology. Further clarification of the underlying chemistry and possible radical pairs involved in the microtubule system has the potential to bridge structural biology, biophysics, and quantum biology, with implications for neurobiology, bioengineering, and medical applications.

MATERIALS AND METHODS

Enriched Mg isotopes

Stable isotopes of ^{25}Mg and ^{26}Mg were purchased in Mg oxide (MgO) form from BuyIsotope (Neonest AB, Solna, Sweden) at >99.38% enrichment levels of isotope to ensure that all effects observed are attributed to specific isotopes and not due to impurities. Natural abundance ^{Nat}MgONatMgO (>99.99%) was purchased from Sigma-Aldrich (529699). MgO was converted to Mg chloride ($MgCl_2$) through a reaction with hydrochloric acid (HCl) and used in tubulin polymerization assays.

Tubulin polymerization assays

Tubulin polymerization assay kits with >99% pure porcine tubulin were purchased from Cytoskeleton Inc. (BK006P). The manufacturer's instructions were followed for the assays. General tubulin buffer from the kits was replaced with in-house made buffer using 80 mM Pipes (pH 6.9; 6910-OP from Millipore), 0.5 mM EGTA (03777 from Sigma-Aldrich), and 2 mM $MgCl_2$ as described above. The eight central wells of a 96-well plate were plated with Mg isotope samples in alternating patterns on each plate with different alternating patterns for each run to allow each isotope equal placement across the plate and between the coils. Tubulin polymerization dynamics were measured via OD_{355} using a Victor NIVO microplate reader (PerkinElmer) following the manufacturer's instructions. Plates were read in the microplate reader for 5 min and then placed in an incubator set at 37°C and ambient lighting containing a GSC

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5-inch Helmholtz coil (fig. S5). After 50 min, the plates were transferred back to the microplate reader and read for another 5 min.

Field measurements

To generate the magnetic field, a GSC 5-inch diameter Helmholtz coil setup, with each coil separated center-to-center by 2.5 inches, was supplied with a 3.6-A current. The magnetic field in each of the eight central wells of a 96-well plate (see fig. S5) were measured in the incubator powered on at 37°C. with a handheld digital high-precision Teslameter (Tunkia TD8620) with the Helmholtz coil powered both on and off. When on, the magnetic field across the wells was measured to be 2.99 ± 0.02 mT, yielding a field homogeneity of ±0.7%. A 3-mT field was chosen as it falls within the characteristic regime identified to affect other biologically plausible RPMs (*55*).

Temperature measurements

Temperatures in each plate well were measured with a Traceable Lollipop waterproof and shockproof thermometer (model 4371).

Statistical analysis

Polymerization data for each replicate were normalized by subtracting the initial OD value for the replicate at time 0 to account for slight timing differences in plating the tubulin solutions. To account for plate-to-plate variations, all polymerization data were then normalized by dividing the time course values by the plate average OD for the last 5 min of the replicates for ^{26}Mg, yielding all measures in ratio to ^{26}Mg. All raw data and normalization procedures are provided in the Supplementary Materials. Statistical comparisons for maximum tubulin polymerization were performed between final endpoints of conditions. Means and SEM were calculated for each condition from their distributions. A two-way analysis of variance (ANOVA) was performed to analyze the effect of Mg isotope, presence of magnetic field, and interaction between the isotope and field. A Tukey-Kramer post hoc analysis was used to account for multiple comparisons. *P* values of less than 0.05 were taken as significant. All statistical analysis was performed in MATLAB 2022a. MATLAB data files and analysis code are provided in the Supplementary Materials.

Tubulin polymerization

Microtubules are dynamic protein polymers essential for eukaryotic cell structure, intracellular transport, and cellular organization (*16*–*18*). Microtubules self-assemble from the αβ-tubulin protein heterodimer through a Mg and GTP hydrolysis–dependent mechanism (Fig. 4A). Free tubulin subunits bind two GTP molecules, with one—the exchangeable GTP on β-tubulin—undergoing hydrolysis to GDP in the presence of a magnesium ion (Mg^{2+}) (*56*, *57*) (Fig. 4B). This conformational change, occurring after tubulin is incorporated into the microtubule, increases the dissociation rate of exposed tubulin subunits at the growing end, thereby regulating the polymerization and depolymerization dynamics of the microtubule (Fig. 4C). As naturally occurring magnesium (^{Nat}Mg) exists in nature as three stable isotopes with varying abundances, ^{24}Mg (79%), ^{25}Mg (10%), and ^{26}Mg (11%), with only ^{25}Mg having a nonzero nuclear spin (5/2+), it is uniquely suited to test spin-dependent isotope effects on GTP hydrolysis via measuring microtubule polymerization.

Model

The chemical equation reads as follows

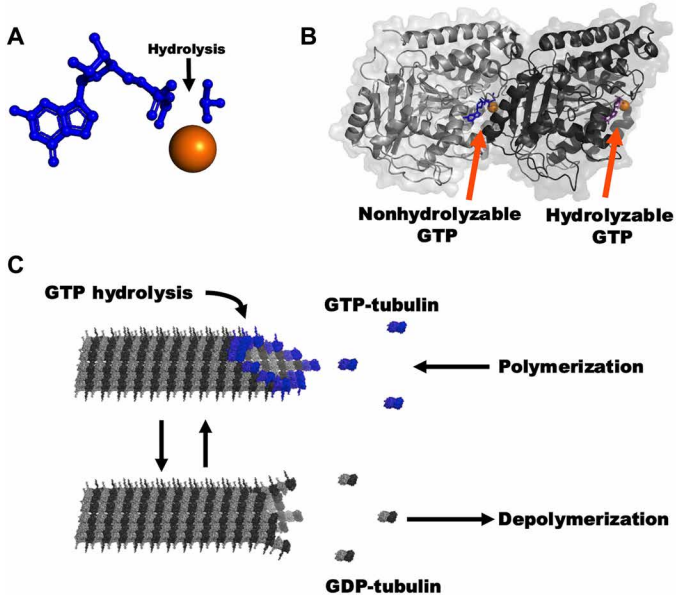


Fig. 4. Tubulin polymerization. (**A**) GTP (blue) hydrolyzes in the presence of a Mg^{2+} ion (orange sphere) to produce GDP. (**B**) Crystal structure of α-tubulin (light gray) and β-tubulin (dark gray) with GTP (blue) and GDP (violet) in the nonhydrolyzable and hydrolyzable sites. (**C**) αβ-tubulin containing GTP (blue tubulin) at the hydrolyzable site polymerizes by attaching to the growing end of a microtubule to form the left-handed, helical, chiral microtubule structure with a hollow center. As the microtubule grows, tubulin hydrolyzes GTP with the aid of a Mg ion, causing the tubulin protein to flex. If the GTP cap on the end of the growing microtubule disappears, the flexed tubulin causes protofilaments to curve and separate from the microtubule, causing the microtubule to depolymerize in catastrophe. If additional GTP tubulin is added, the microtubule is rescued and continues to grow. Modified image from (*32*) under a Creative Commons Attribution License.

$$\frac{d[MT(t)]}{dt} = k_p[Tu(t)] - k_d[MT(t)] \tag{3}$$

which yields $[MT(t)] = \frac{k_p[P]}{k_p + k_d}[1 - e^{-(k_p+k_d)t}]$.

The state of the radical pair is characterized by the spin density operator, which encapsulates the system's quantum state. The time evolution of the spin density matrix is influenced by a combination of coherent spin dynamics, chemical reactivity, and spin relaxation processes. This evolution is governed by the Liouville master equation, which provides the time dependence of the spin density operator for the radical pair system

$$\frac{d\hat{\rho}(t)}{dt} = -\hat{\hat{L}}[\hat{\rho}(t)] \tag{4}$$

Here, the Liouvillian superoperator is expressed as $\hat{\hat{L}} = i\hat{\hat{H}} + \hat{\hat{K}} + \hat{\hat{R}}$, where $\hat{\hat{H}}$, $\hat{\hat{K}}$, and $\hat{\hat{R}}$ represent the Hamiltonian superoperator, the chemical reaction superoperator, and the spin relaxation superoperator, respectively. These components collectively determine the dynamics of the spin density matrix. Further details about these terms can be found in the Supplementary Materials.

Beyond the conventional singlet and triplet states, radical pairs can also originate as F-pairs, which arise when two radicals form independently rather than through direct electron transfer between them. In this case, the initial state is a completely mixed state,

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represented as $\frac{1}{4M}\hat{I}_{4M}$, where $\hat{I}_{4M}$ is the identity matrix of dimension $4M$ (*33*). M is the nuclear spin multiplicity.

The Hamiltonian is expressed as follows

$$\hat{H} = \omega\hat{S}_{A_z} + a_A\hat{\mathbf{S}}_A.\hat{\mathbf{I}}_A + \omega\hat{S}_{B_z} + a_B\hat{\mathbf{S}}_B.\hat{\mathbf{I}}_B + a_{Mg}\hat{\mathbf{S}}_B.\hat{\mathbf{I}}_{Mg} \quad (5)$$

Here, $\hat{\mathbf{S}}_A$ and $\hat{\mathbf{S}}_B$ represent the electron spin operators for the radical pairs labeled A and B, respectively. The Larmor frequency of the electrons, resulting from the Zeeman interaction, is denoted by ω. In addition, $\hat{\mathbf{I}}_A$ and $\hat{\mathbf{I}}_B$ correspond to the nuclear spin operator coupled with electron A, and B, respectively, whereas $\hat{\mathbf{I}}_{Mg}$ represents the nuclear spin operator of ^{25}Mg coupled to electron B. The parameter a_A and a_B refers to the HFCC of nuclear spins coupled to electron A and B, respectively. a_{Mg} denotes the HFCC of ^{25}Mg.

For spin-selective chemical reactions, we use the Haberkorn superoperator (*11*, *58*), which is given by the following equation

$$|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (6)$$

$$|T_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), |T_{+1}\rangle = |\uparrow\uparrow\rangle, |T_{-1}\rangle = |\downarrow\downarrow\rangle \quad (7)$$

$$\hat{\hat{K}} = \frac{1}{2}k_S\left(\hat{P}^S \otimes I_{4M} + I_{4M} \otimes \hat{P}^S\right) + \frac{1}{2}k_T\left(\hat{P}^T \otimes I_{4M} + I_{4M} \otimes \hat{P}^T\right) \quad (8)$$

where $\hat{P}^S$ and $\hat{P}^T$, respectively, are the singlet and triplet projection operators

$$\frac{1}{M}\hat{P}^S = |S\rangle\langle S| \otimes \frac{1}{M}\hat{I}_M \quad (9)$$

$$\frac{1}{3M}\hat{P}^T = \frac{1}{3}\left(|T_0\rangle\langle T_0| + |T_{+1}\rangle\langle T_{+1}| + |T_{-1}\rangle\langle T_{-1}|\right) \otimes \frac{1}{M}\hat{I}_M \quad (10)$$

Spin relaxation is modeled via random time-dependent local fields (*59*, *60*), and the corresponding superoperator reads as follows

$$\hat{\hat{R}} = r_A\left[\frac{3}{4}I_{4M} \otimes I_{4M} - \hat{S}_{A_x} \otimes \left(\hat{S}_{A_x}\right)^T - \hat{S}_{A_y} \otimes \left(\hat{S}_{A_y}\right)^T - \hat{S}_{A_z} \otimes \left(\hat{S}_{A_z}\right)^T\right] + r_B\left[\frac{3}{4}I_{4M} \otimes I_{4M} - \hat{S}_{B_x} \otimes \left(\hat{S}_{B_x}\right)^T - \hat{S}_{B_y} \otimes \left(\hat{S}_{B_y}\right)^T - \hat{S}_{B_z} \otimes \left(\hat{S}_{B_z}\right)^T\right] \quad (11)$$

where the symbols have the abovementioned meanings. The ultimate fractional triplet yield for F-pairs, for time periods significantly exceeding the radical pair lifetime, is expressed as

$$\Phi_T = k_T Tr\left[\hat{P}^T\hat{\hat{L}}^{-1}\left(\frac{\hat{I}_{4M}}{4M}\right)\right] \quad (12)$$

Supplementary Materials

The PDF file includes:

Figs. S1 to S5

Legends for data S1 to S6

Other Supplementary Material for this manuscript includes the following:

Data S1 to S6

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